

NeSS M2023

HW1: Circuits

Q1. • ANTI-THEFT ALARM (vehicles)

Resistors and capacitors are used for voltage regulation; diodes are used too and the whole circuit takes input from the ignition switch and outputs the alarm signal.

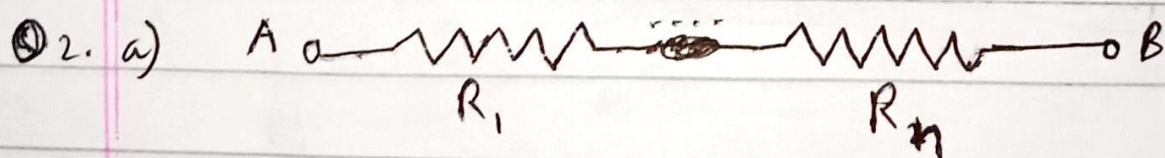
• AUTOMATIC SLIDING DOORS

Uses a PIR (motion) detector, whose input to a circuit of resistors, grounding capacitors and transistors drive the door system.

Contactless

• THERMAL GUN (~~Frontal~~ temperature)

Takes infrared sensory input, processes it through a circuit, and gives an output to the display.



Let the (common) current be i .

$$\Rightarrow V_A - iR_1 - iR_2 - \dots - iR_n = V_B$$

[via Kirchhoff's Voltage Law]

$$\Rightarrow V_A - V_B = i(R_1 + R_2 + \dots + R_n)$$

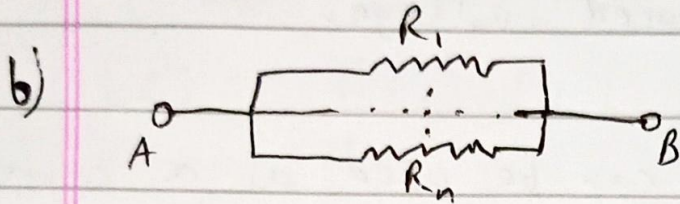
$$\Rightarrow \frac{V_A - V_B}{i} = \sum_{k=1}^n R_k = \frac{\Delta V}{i} \quad \text{--- (1)}$$

(2)

Eq. circuit: $A \xrightarrow{R_{eq}} i \rightarrow B$

$$\Rightarrow R_{eq} = \frac{V_A - V_B}{i} = \frac{\Delta V}{i} \quad - (2)$$

Comparing (1) & (2), $\boxed{R_{eq} = \sum_{k=1}^n R_k}$



Let the currents in each branch be $i_1, i_2, \dots, i_n$

$$\Rightarrow V_A - i_1 R_1 = V_B$$

$$\Rightarrow V_A - i_k R_k = V_B, \quad 1 \leq k \leq n,$$

$$\Rightarrow \frac{V_A - V_B}{R_k} = i_k, \quad 1 \leq k \leq n.$$

$$\Rightarrow \sum_{k=1}^n i_k = \sum_{k=1}^n \frac{V_A - V_B}{R_k} = i \text{ (net current)} \quad - (4)$$

Eq. circuit: $A \xrightarrow{R_{eq}} i \rightarrow B$

$$\Rightarrow R_{eq} = \frac{V_A - V_B}{i} \quad - (3)$$

By (4), $\frac{V_A - V_B}{i} = \sum_{k=1}^n \frac{1}{R_k} = \frac{i}{V_A - V_B}$

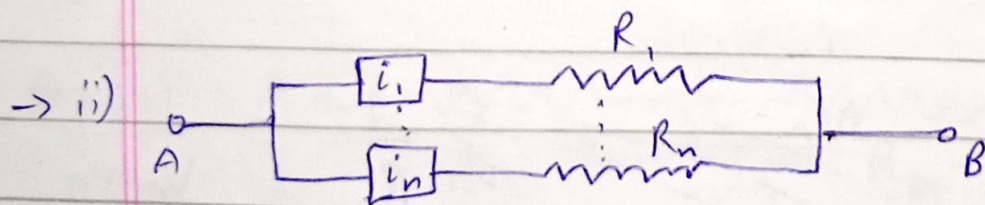
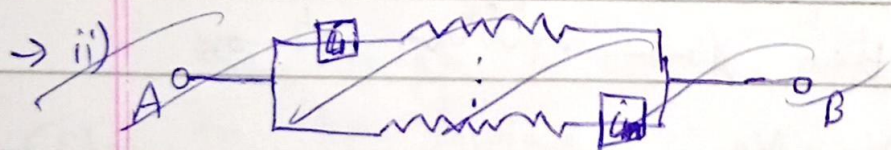
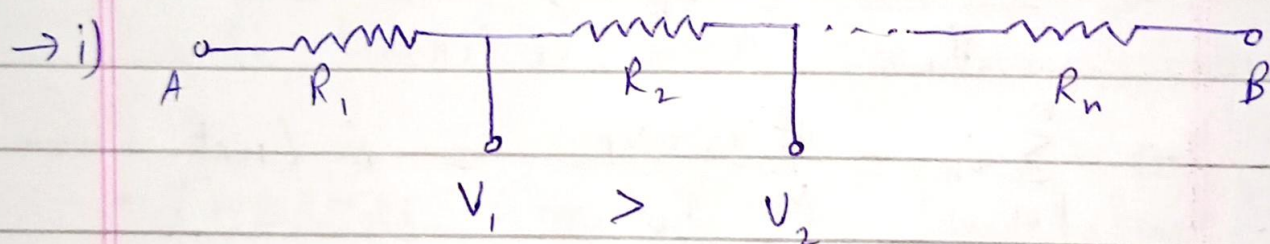
By comparison,

$$\cancel{R_{eq}} = \sum_{k=1}^n$$

$$\boxed{\frac{1}{R_{eq}} = \sum_{k=1}^n \frac{1}{R_k}}$$

i) Combination (a) can be used as a voltage divider because ~~it~~ ~~see~~ the voltage drops across each resistor, so at the second terminal of every resistor one can obtain a reduced voltage.

ii) Combination (b) can be used as a current divider, since the current through each branched resistor can be controlled by changing resistor values. Higher the resistance, lesser the current through its branch.



Q 3. $R = \frac{V}{I} = \frac{\text{Volts}}{\text{Amps}} = \text{Ohms. } (\Omega)$

$$[R] = \frac{[V]}{[I]} = \frac{[M L^2 T^{-2}]}{[A] \cdot [AT]} = [M L^2 T^{-3} A^{-2}]$$

$$\Rightarrow 1 \Omega = 1 \text{ kg m}^2 \text{ s}^{-3} \text{ A}^{-2}$$

$L \rightarrow$ Henry (H)

$$[L] = \frac{[MLT^{-2}] \cdot [L^2]}{[A^2] \cdot [L]} = [ML^2 T^{-2} A^{-2}]$$

$$\Rightarrow 1 H = 1 \text{ kg m}^2 \text{ s}^{-2} \text{ A}^{-2}$$

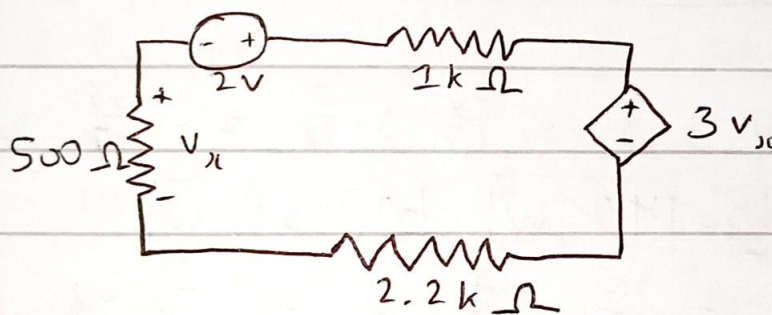
$C \rightarrow$ Farad (F)

$$[C] = [\epsilon_0] \cdot [L] = [M^{-1} L^{-2} T^4 A^2] [L] \\ = [M^{-1} L^{-1} T^4 A^2]$$

$$\Rightarrow 1 F = 1 \text{ kg}^{-1} \text{ m}^{-2} \text{ s}^4 \text{ A}^2$$

Q4. ECA Ch-3.

27)



Let the current be i .

Using KVL, $2 - 1ki - 3V_x - 2.2ki - 500i = 0$

Also, $V_x = -500i$.

$$\Rightarrow i(1000 + 2200 + 500 - 3 \cdot 500) = 2$$

$$\Rightarrow i = \frac{2}{3700 - 1500} = \frac{2}{2200} \text{ A}$$

$$\Rightarrow i = 0.91 \text{ mA}$$

$$\therefore P_{1k\Omega} = i^2(1k) = 0.826 \text{ mW}$$

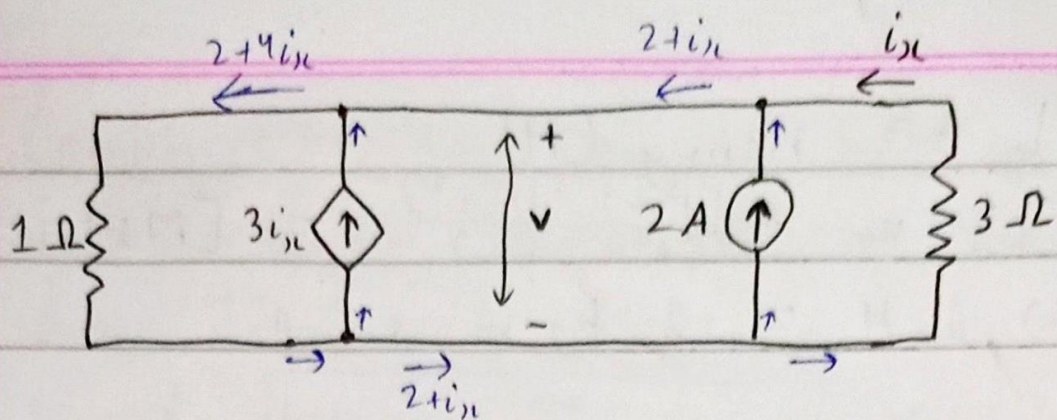
$$P_{3V_x} = 3V_x \cdot i = 1500 i^2 = 1.239 \text{ mW}$$

$$P_{2.2k\Omega} = i^2(2.2k) = 1.818 \text{ mW}$$

$$P_{500\Omega} = i^2(500) = 0.413 \text{ mW}$$

$$P_{2V} = -2i = -1.818 \text{ mW} \quad (\text{Power is supplied})$$

33)



[Current through wires via KCL]

Using KVL across the outermost loop:

$$-3i_x - 1(2+4i_x) = 0$$

$$\Rightarrow 7i_x = -2 \Rightarrow i_x = -2/7 \text{ A}$$

$$\therefore v = -3 \cdot i_x = -3(-2/7) = 6/7 \text{ V}$$

$$\Rightarrow \boxed{v = 0.857 \text{ V}}$$

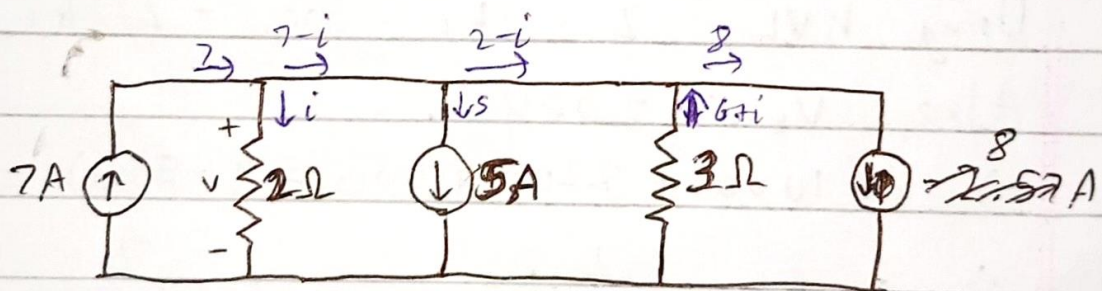
$$P_{3i_x} = v \cdot 3i_x = \frac{6}{7} \cdot 3 \cdot \frac{-2}{7} = -\frac{36}{49} \text{ W}$$

$$= \boxed{-0.734 \text{ W}}$$

$$P_{2A} = v \cdot 2 = \frac{6}{7} \cdot 2 = \frac{12}{7} \text{ W}$$

$$= \boxed{1.714 \text{ W}}$$

39)



a) Let i be the current through 2Ω .

Using KVL across $2\Omega - 3\Omega$ loop:

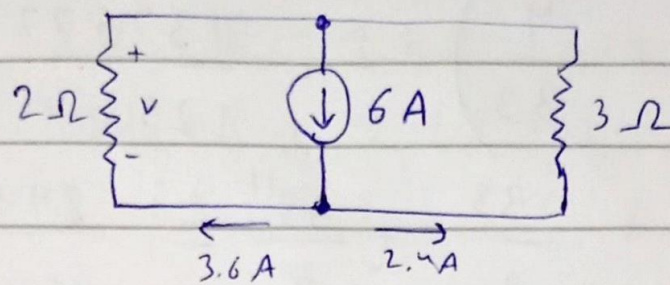
$$-2i - 3(6+i) = 0 \Rightarrow 18 + 5i = 0$$

$$\Rightarrow i = -18/5 \text{ A}$$

$$\Rightarrow i = \underbrace{-3.6 \text{ A}}_{-i_{2\Omega}} ; i+6 = \underbrace{2.4 \text{ A}}_{i_{3\Omega}}$$

$$\Rightarrow i_{\text{net}} = i_{2\Omega} + i_{3\Omega} = 6 \text{ A}$$

(6)



$$\therefore V = -2 \cdot (3.6) = \boxed{-7.2 \text{ V}}$$

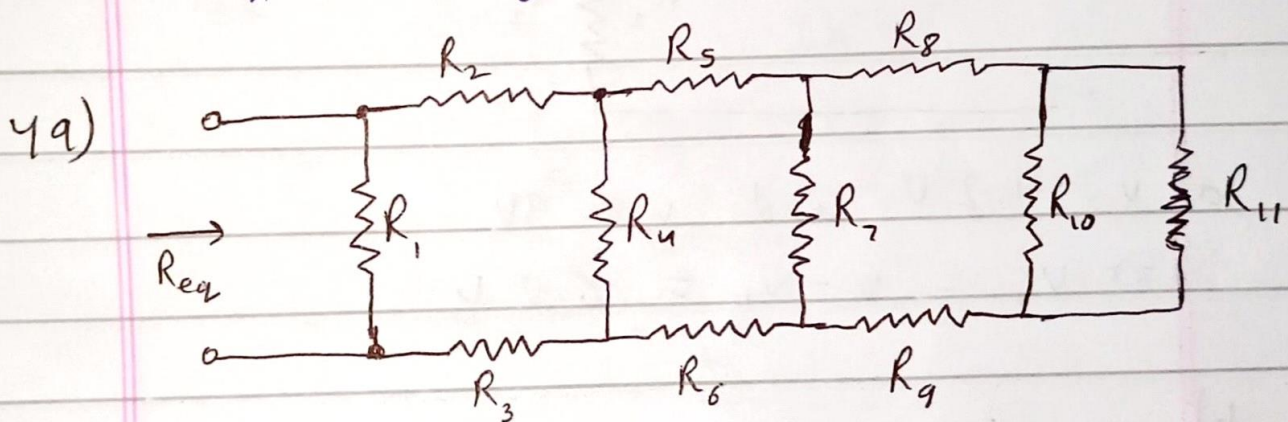
$$b) P_{6A} = 7.2 \times 6 = 43.2 \text{ W}$$

$$P_{7A} = 7V = -50.4 \text{ W}$$

$$P_{5A} = -5V = 36 \text{ W}$$

$$P_{8A} = -8V = 57.6 \text{ W}$$

$$\Rightarrow P_{7A} + P_{5A} + P_{8A} = 43.2 \text{ W} = P_{6A} \quad \checkmark$$



$$R_1 = 2R_2 = 3R_3 = 4R_4 = \dots ; R_{11} = 3 \Omega$$

$$\left(\frac{1}{R_{10}} + \frac{1}{R_{11}} \right)^{-1} = \left(\frac{11}{33} + \frac{10}{33} \right)^{-1} = \frac{33}{21} \Omega$$

$$\frac{33}{21} + R_8 + R_9 = \frac{33}{21} + \frac{33}{8} + \frac{33}{9} = \frac{4719}{504} \Omega$$

$$\left(\frac{504}{4719} + \frac{1}{R_7} \right)^{-1} = \frac{504}{4719} \left(\frac{168}{1573} + \frac{7}{33} \right)^{-1}$$

$$= \left(\frac{1505}{4719} \right)^{-1} = \frac{4719}{1505} \Omega$$

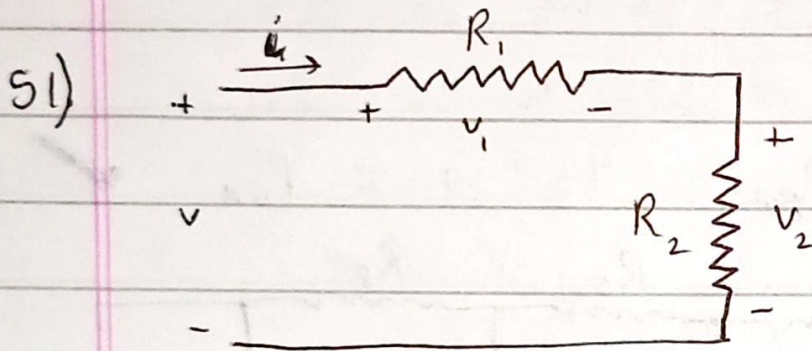
$$\frac{4719}{1505} + R_5 + R_6 = \frac{4719}{1505} + \frac{33}{5} + \frac{33}{6} = \frac{13757}{9030} \Omega$$

$$\left(\frac{9030}{137577} + \frac{1}{33} \right)^{-1} = \frac{137577}{25706} \Omega$$

$$\frac{137577}{25706} + \frac{33}{2} + \frac{33^{11}}{3} = \frac{844492}{25706} \Omega$$

$$\left(\frac{25706}{844492} + \frac{1}{33} \right)^{-1} = \frac{2533476}{153890} \Omega$$

$$\therefore R_{eq} = 16.463 \Omega$$



a) $v = 9.2 \text{ V}$ and $v_1 = 3 \text{ V}$
 $\Rightarrow v_2 = v - v_1 = 6.2 \text{ V}$

b) $v_2 = 1 \text{ V}$ and $v = 2 \text{ V}$
 $\Rightarrow v_1 = v - v_2 = 1 \text{ V}$

c) $v_1 = 3 \text{ V}$ and $v_2 = 6 \text{ V}$
 $\Rightarrow v = v_1 + v_2 = 9 \text{ V}$

d) $v_1 = v_2$
 $\Rightarrow iR_1 = iR_2 \Rightarrow R_1/R_2 = 1$

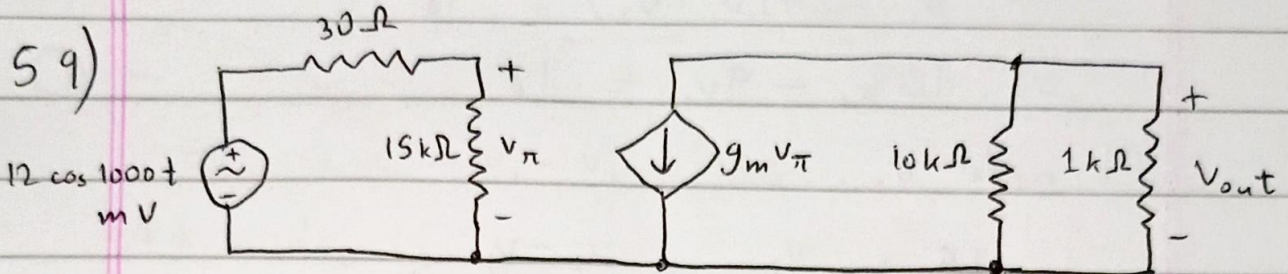
e) $v = 3.5 \text{ V}$ and $R_1 = 2R_2$
 $\Rightarrow v = i(R_1 + R_2) = i(3R_2) = 3.5 \text{ V} \Rightarrow v_2 = iR_2 = \underline{\underline{1.167 \text{ V}}}$

(8)

f) $v = 1.8 \text{ V}$, $R_1 = 1 \text{ k}\Omega$ and $R_2 = 4.7 \text{ k}\Omega$

$$\Rightarrow v = i(R_1 + R_2) \Rightarrow 1.8 = i \cdot 5700 \Rightarrow i = \frac{1.8}{5700}$$

$$\Rightarrow V_1 = i R_1 = \frac{1.8}{5700} \times 1000 = \frac{1.8}{5.7} = \underline{\underline{0.315 \text{ V}}}$$



$$g_m = 1.2 \text{ mS}$$

$$v_\pi = \frac{(5 \text{ k}\Omega)(12 \cos 1000t \text{ mV})}{15030} \approx 12 \cos 1000t \text{ mV}$$

$$i_{1 \text{ k}\Omega} = \frac{g_m v_\pi}{\left(\frac{1}{10 \text{ k}\Omega} + \frac{1}{1 \text{ k}\Omega}\right)^{-1}} = \frac{g_m v_\pi}{1 \text{ k}\Omega}$$

$$= \frac{1.2 \times 10^{-3} \times 12 \cos 1000t \text{ mV} \times 10 \times 10^{-3}}{10^3 \times 11} \text{ A}$$

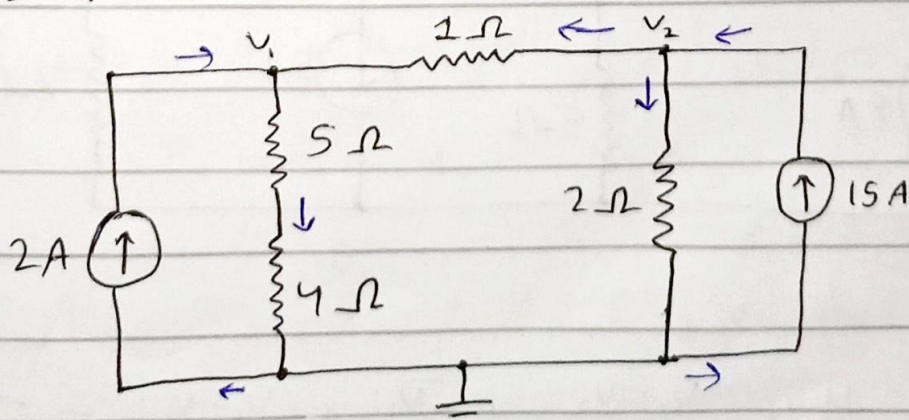
$$= \frac{1.2 \times 10^{-3} \times 12 \cos 1000t \text{ mV} \times 10 \times 10^{-3}}{10^3 \times 11} \text{ A}$$

$$\approx \boxed{13.091 \cos 1000t \text{ }\mu\text{A}}$$

$$\therefore V_{out} = -i_{1 \text{ k}\Omega} \cdot 1 \text{ k}\Omega = \underline{\underline{-13.091 \cos 1000t \text{ mV}}}$$

Q5. ECA Ch-4.

10)



At node v_1 :

$$-2 + \frac{v_1}{9\Omega} + \frac{v_1 - v_2}{1\Omega} = 0$$

$$\Rightarrow v_1 + 9(v_1 - v_2) = 18$$

$$\Rightarrow 10v_1 - 9v_2 = 18 \quad - (1)$$

At node v_2 :

$$-15 + \frac{v_2}{2\Omega} + \frac{v_2 - v_1}{1\Omega} = 0$$

$$\Rightarrow v_2 + 2(v_2 - v_1) = 15 \times 2 = 30$$

$$\Rightarrow 3v_2 - 2v_1 = 30 \quad - (2)$$

$$(1) + 3 \times (2) :-$$

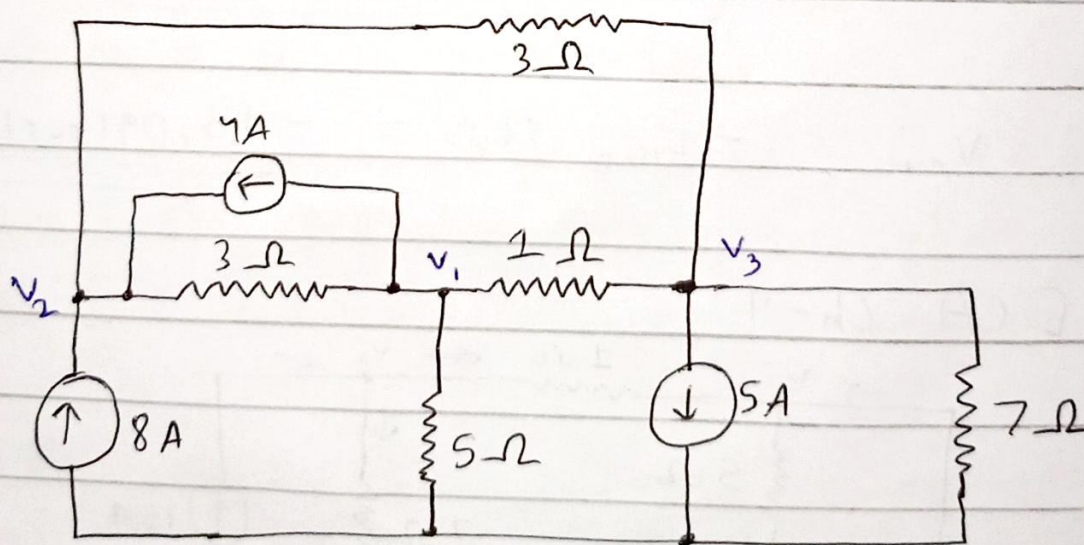
$$10v_1 - 9v_2 + 9v_2 - 6v_1 = 18 + 90$$

$$\Rightarrow 4v_1 = 108 \Rightarrow v_1 = 27 \text{ V}$$

$$\text{From (2) :- } v_2 = \frac{1}{3}(30 + 2v_1) = 28 \text{ V}$$

$$\therefore \boxed{v_1 - v_2 = -1 \text{ V}}$$

13)



At node v_1 :

$$4 + \frac{v_1 - v_2}{3} + \frac{v_1}{5} + \frac{v_1 - v_3}{1} = 0$$

(10)

$$\Rightarrow 60 + 5(v_1 - v_2) + 3v_1 + 15(v_1 - v_3) = 0$$

$$\Rightarrow 23v_1 - 5v_2 - 15v_3 + 60 = 0 \quad - (1)$$

At node v_2 :

$$-8 - 4 + \frac{v_2 - v_1}{3} + \frac{v_2 - v_3}{3} = 0$$

$$\Rightarrow 2v_2 - v_1 - v_3 = 36 \quad - (2)$$

At node v_3 :

$$5 + \frac{v_3 - v_1}{1} + \frac{v_3}{7} + \frac{v_3 - v_2}{3} = 0$$

$$\Rightarrow 105 + 21(v_3 - v_1) + 3v_3 + 7(v_3 - v_2) = 0$$

$$\Rightarrow 31v_3 - 21v_1 - 7v_2 + 105 = 0 \quad - (3)$$

From (2) :- $v_1 = 2v_2 - v_3 - 36$

Using this in (3) :- $31v_3 - 21(2v_2 - v_3 - 36) - 7v_2 + 105 = 0$

$$\Rightarrow 52v_3 - 49v_2 + 861 = 0$$

~~And in~~ $\Rightarrow v_3 = \frac{1}{52}(861 - 49v_2) = \frac{1}{52}(49v_2 - 861)$

And in (1) :- $23(2v_2 - v_3 - 36) - 5v_2 - 15v_3 + 60 = 0$

$$\Rightarrow 41v_2 - 38v_3 - 768 = 0$$

$$\Rightarrow 41v_2 - 38 \cdot \frac{1}{52}(49v_2 - 861) - 768 = 0$$

$$\Rightarrow 270v_2 - 7218 = 0 \Rightarrow v_2 = \frac{3609}{135} = \frac{401}{15}$$

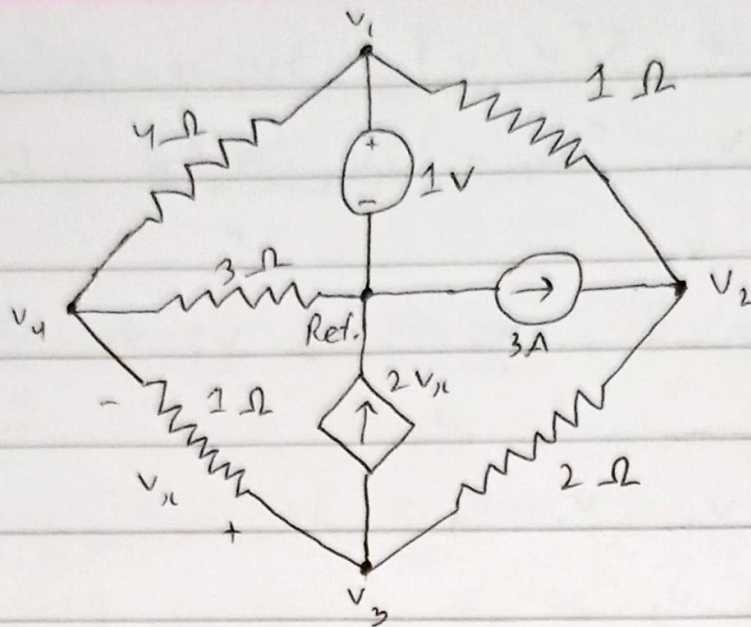
$$\Rightarrow v_3 = (41v_2 - 768) \frac{1}{38} = \frac{4921}{15} \times \frac{1}{38} = \frac{259}{30}$$

$$\therefore v_1 = 2v_2 - v_3 - 36 = 2\left(\frac{401}{15}\right) - \frac{259}{30} - 36$$

$$= \frac{265}{30} = \frac{53}{6} \Rightarrow \boxed{v_1 = 8.83 \text{ V}}$$

$$\therefore P_{7\Omega} = \frac{v_3^2}{7} = 10.647 \text{ W}$$

28)



$$\boxed{v_1 = 1V}$$

At node v_2 :

$$\frac{v_2 - v_1}{1} - 3 + \frac{v_2 - v_3}{2} = 0$$

$$\Rightarrow 2(v_2 - v_1) - 6 + v_2 - v_3 = 0$$

$$\Rightarrow 3v_2 - 2v_1 - v_3 = 6 \quad - (1)$$

$$\Rightarrow v_3 = 3v_2 - 2v_1 - 6 = 3v_2 - 8 \quad (v_1 = 1V)$$

At node v_3 :

$$\frac{v_3 - v_2}{2} + 2v_x + \frac{v_3 - v_4}{1} = 0$$

$$\Rightarrow v_3 - v_2 + 4(v_3 - v_4) + 2(v_3 - v_4) = 0$$

$$\Rightarrow 7v_3 - v_2 - 6v_4 = 0$$

$$\Rightarrow 7(3v_2 - 2v_1 - 6) - v_2 - 6v_4 = 0 \quad (1)$$

$$\Rightarrow 20v_2 - 14v_1 - 6v_4 - 42 = 0$$

$$\Rightarrow 20v_2 - 6v_4 - 56 = 0 \quad (v_1 = 1V) \quad - (2)$$

$$\Rightarrow v_4 = \frac{1}{6}(20v_2 - 56) = \frac{1}{3}(10v_2 - 28)$$

At node v_4 :

$$\frac{v_4 - v_3}{1} + \frac{v_4}{3} + \frac{v_4 - 1}{4} = 0$$

(12)

$$\Rightarrow 12(v_4 - v_3) + 9v_4 + 3(v_4 - 1) = 0$$

$$\Rightarrow 19v_4 - 12v_3 - 3 = 0$$

Using ① & ② :-

$$19(10v_2 - 28) - 12(3v_2 - 8) = 0$$

$$\Rightarrow 190 - 36 - 81v_2 - 244 = 0$$

$$\Rightarrow v_2 = \frac{244}{81} \text{ V}$$

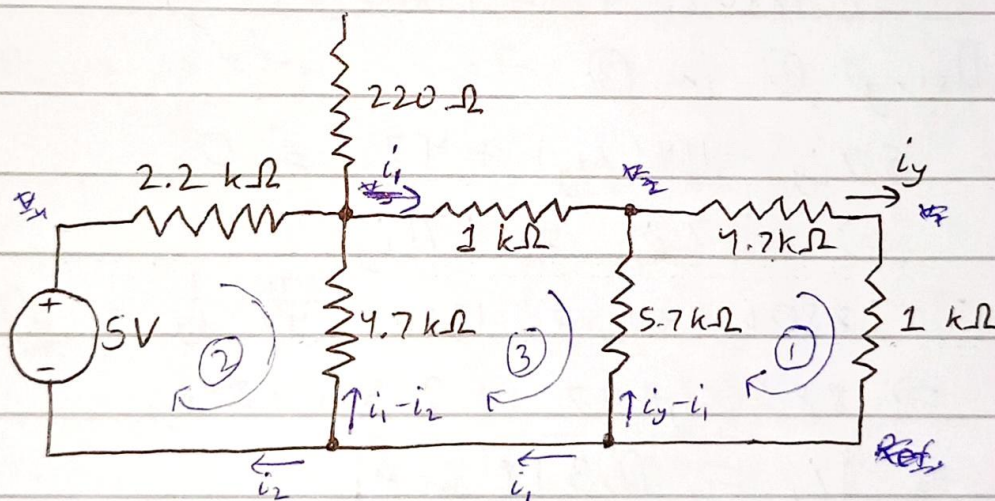
$$\Rightarrow v_3 = 3v_2 - 8 = \frac{244}{27} - 8 = \frac{28}{27} \text{ V} \quad (10)$$

$$\Rightarrow v_4 = \frac{1}{3} \left(10 \cdot \frac{244}{81} - 28 \right) = \frac{172}{243} \text{ V}$$

$$\therefore \boxed{v_1 = 1 \text{ V}} ; \boxed{v_2 = 3.01 \text{ V}}$$

$$\boxed{v_3 = 1.03 \text{ V}} ; \boxed{v_4 = 0.7 \text{ V}}$$

34)



a) Node 1: $\frac{v_1}{1020} + \frac{v_1 - v_2}{4700} = 0$

$$\Rightarrow 47v_1 + 10(v_1 - v_2) = 57v_1 - 10v_2 = 0$$

Node 2:

$$\frac{v_2 - v_1}{4700} + \frac{v_2}{5700} + \frac{v_2 - v_3}{1000} = 0$$

$$\Rightarrow 570(v_2 - v_1) + 470v_2 + 2679(v_2 - v_3) = 0$$

$$\Rightarrow 3719v_2 - 570v_1 - 2679v_3 = 0$$

~~Node 325~~ KVL at loop 1:

$$-4.7k\Omega - 4700i_y - 1000i_y - 5700(i_y - i_1) = 0$$

$$\Rightarrow 114i_y = 57i_1 \Rightarrow \underline{i_1 = 2i_y} \quad - (1)$$

KVL at loop 2:

$$-2200i_2 + 4700(i_1 - i_2) + 5 = 0$$

$$\Rightarrow 4700i_1 + 5 = 6900i_2$$

$$\Rightarrow 940i_1 + 1 = 1380i_2 \quad - (2)$$

KVL @ loop 3:

$$-1000i_1 + 5700(i_y - i_1) - 4700(i_1 - i_2) = 0$$

$$\Rightarrow 57i_y - 114i_1 + 47i_2 = 0 \quad - (3)$$

Using (1) in (2) :-

$$940(2i_y) + 1 = 1380i_2$$

$$\Rightarrow 1880i_y + 1 = 1380i_2$$

Using (1) in (3) :-

$$57i_y - 114(2i_y) + 47i_2 = 0$$

$$\Rightarrow 47i_2 = 171i_y$$

$$\therefore 1880i_y + 1 = 1380 \cdot \frac{171}{47} \cdot i_y$$

$$\Rightarrow 88360i_y + 47 = 235980i_y$$

$$\Rightarrow \boxed{i_y = 0.318 \text{ mA}}$$

$$i_{220\Omega} = 0 \text{ A} \Rightarrow \boxed{P_{220\Omega} = 0 \text{ W}}$$

$$4) a) \quad 3 = \frac{v_1}{5} - \frac{v_2 - v_1}{22} + \frac{v_1 - v_3}{3} \quad - (1)$$

$$2 - 1 = \frac{v_2 - v_1}{22} + \frac{v_2 - v_3}{14} \quad - (2)$$

$$0 = \frac{v_3}{10} + \frac{v_2 - v_1}{3} + \frac{v_3 - v_2}{14} \quad - (3)$$

$$\textcircled{1}:- 990 = 66v_1 - 15(v_2 - v_1) + 110(v_1 - v_3)$$

$$\Rightarrow 191v_1 - 15v_2 - 110v_3 = 990$$

$$\textcircled{2}:- 154 = 7(v_2 - v_1) + 11(v_2 - v_3)$$

$$\Rightarrow 18v_2 - 7v_1 - 11v_3 = 154$$

$$\textcircled{3}:- 0 = 42v_3 + 140(v_3 - v_1) + 30(v_3 - v_2)$$

$$\Rightarrow \overset{106}{21}v_3 - \overset{140}{70}v_1 - \overset{15}{30}v_2 = 0$$

$$D = \begin{vmatrix} 191 & -15 & -110 \\ -7 & 18 & -11 \\ -70 & -15 & 106 \end{vmatrix} = 160083$$

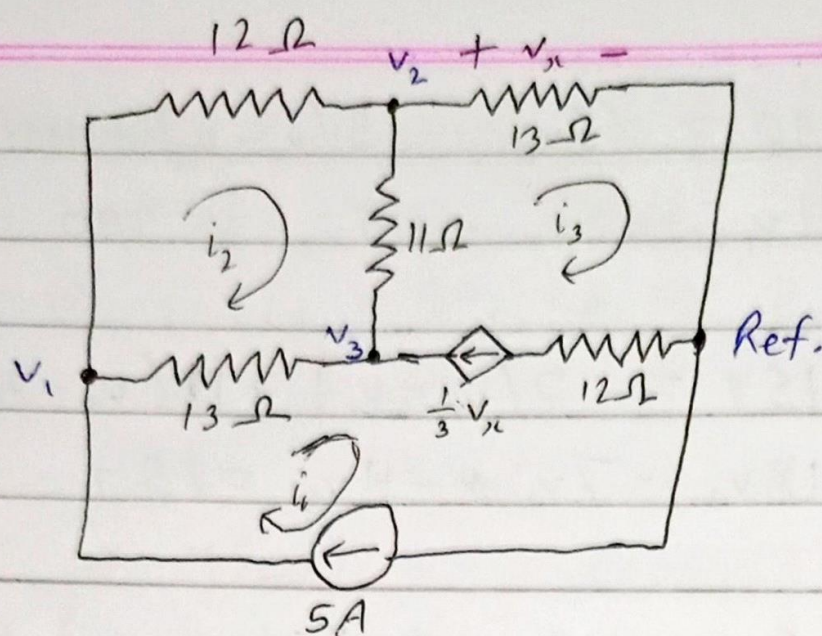
$$D_1 = \begin{vmatrix} 990 & -15 & -110 \\ 154 & 18 & -11 \\ 0 & -15 & 106 \end{vmatrix} = 2224530$$

$$D_2 = \begin{vmatrix} 191 & 990 & -110 \\ -7 & 154 & -11 \\ -70 & 0 & 106 \end{vmatrix} = 3428964$$

$$D_3 = \begin{vmatrix} 191 & -15 & 990 \\ -7 & 18 & 154 \\ -70 & -15 & 0 \end{vmatrix} = 1954260$$

$$\therefore \begin{cases} v_1 = D_1/D = 1070/77 \\ v_2 = D_2/D = 4948/231 \\ v_3 = D_3/D = 940/77 \end{cases}$$

Q6.



KCL: Node 1 :- $5 + \frac{v_3 - v_1}{13} = \frac{v_1 - v_2}{12}$

Node 2 :- $\frac{v_1 - v_2}{12} + \frac{v_3 - v_2}{11} = \frac{v_2}{13}$

Node 3 :- $\frac{v_3 - v_1}{13} + \frac{v_3 - v_2}{11} = \frac{v_2}{3} \quad [V_x = v_2 - 0]$

$\therefore v_1 = \underline{-3.33 \text{ V}} ; v_2 = \underline{-14.5 \text{ V}} ; v_3 = \underline{-50.819 \text{ V}}$

KVL: $i_1 = \underline{5 \text{ A}} ; i_3 = i_1 = \frac{1}{3} V_x = \frac{13}{3} i_3$

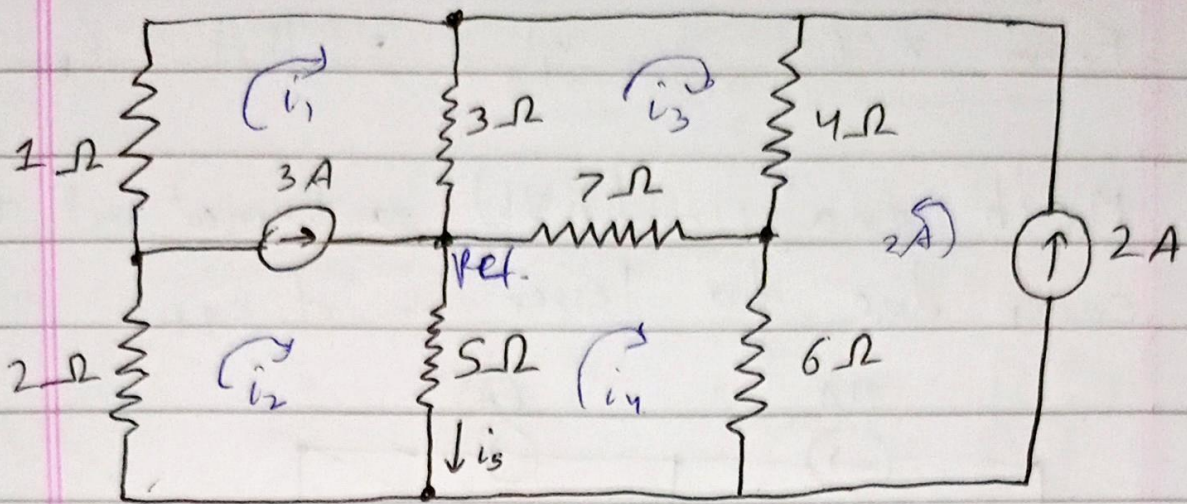
$\Rightarrow i_3 = \underline{-1.5 \text{ A}}$

$12i_2 + 11(i_2 - i_3) + 13(i_2 - i_1) = 0$

$\Rightarrow i_2 = \underline{1.347 \text{ A}}$

From KCL, $\frac{v_1 - v_2}{12} = 1.3475 \text{ A} = i_2 \quad \checkmark$

Here, KVL resulted in only two equations (1 of 3 reduced to single-variable) to solve, so it was simpler.

KVL

$$4i_1 + 7i_2 - 3i_3 - 5i_4 = 0$$

$$3i_1 - 14i_3 + 7i_4 = 8$$

$$5i_2 + 7i_3 - 18i_4 = 12$$

$$\Rightarrow i_1 = \underline{-3.11 \text{ A}} \quad ; \quad i_2 = \underline{-0.11 \text{ A}}$$

$$i_3 = \underline{-1.97 \text{ A}} \quad ; \quad i_4 = \underline{-1.76 \text{ A}}$$

KCL

$$\frac{v_4 - v_1}{1} + \frac{0 - v_1}{3} + \frac{v_3 - v_1}{4} + 2 = 0$$

$$\frac{v_4 - v_2}{2} + \frac{0 - v_2}{5} + \frac{v_5 - v_2}{6} - 2 = 0$$

$$\frac{0 - v_3}{7} + \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{6} = 0$$

$$\frac{v_1 - v_4}{1} + \frac{v_2 - v_4}{2} - 3 = 0$$

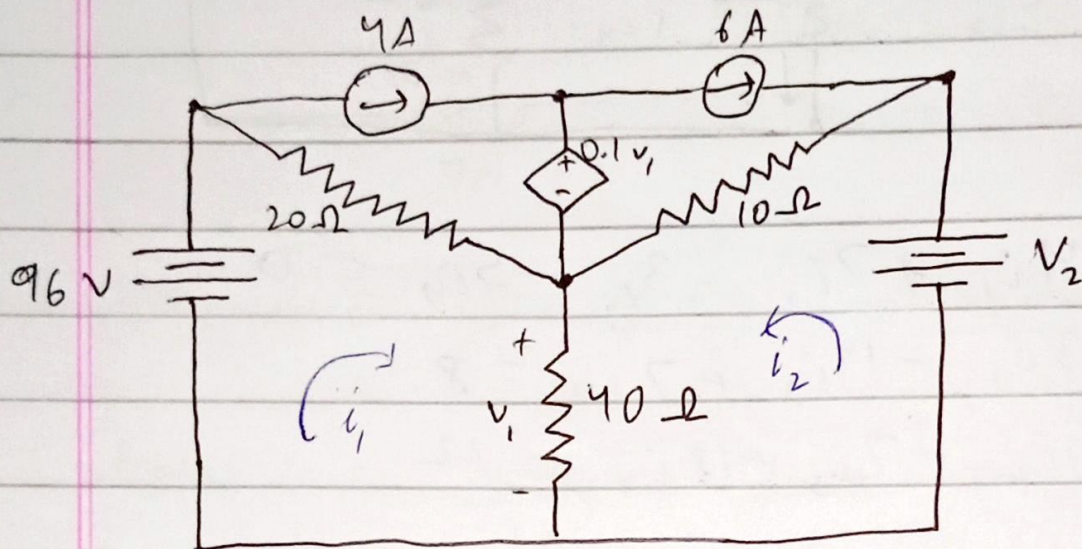
$$\Rightarrow v_1 = \underline{-3.72 \text{ V}} \quad ; \quad v_2 = \underline{-6.76 \text{ V}}$$

$$; \quad v_3 = \underline{-3.54 \text{ V}} \quad ; \quad v_4 = \underline{6.53 \text{ V}}$$

(17)

From KCL, $v_4 - v_1 = -3.4 A = i_1$ ✓

Mesh analysis (KVL) ~~can~~ turned out to be easier due to lesser no. of eqs.



KCL

$$6 + \frac{v_1 - v_2}{10} - 4 + \frac{v_1 - 96}{20} + \frac{v_1}{20} = 0$$

$$\Rightarrow v_1 = \frac{4v_2 + 112}{7} = \left(\frac{4}{7}v_2 + 16 \right) V$$

KVL

$$96 - 20(i_1 - 4) - 40(i_1 - i_2) = 0$$

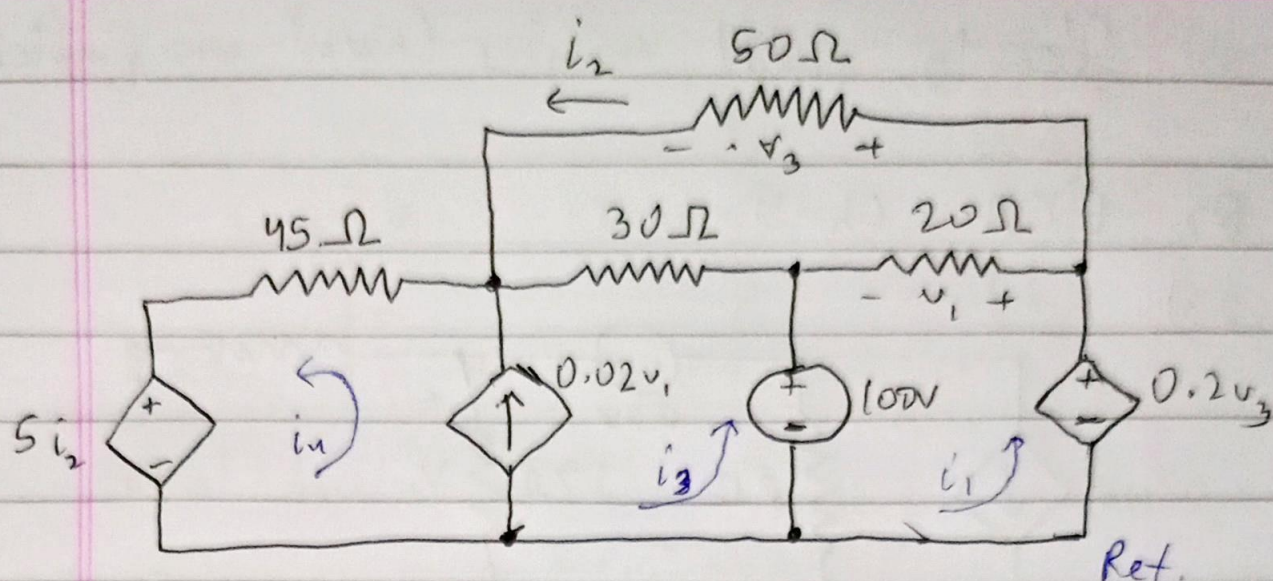
$$\Rightarrow 15i_1 - 10i_2 = 44 \quad - (1)$$

$$v_2 - 10(6 - i_2) - 40(i_1 - i_2) = 0$$

$$\Rightarrow 40i_1 - 50i_2 = v_2 - 60 \quad - (2)$$

$$\Rightarrow i_1 = \frac{280 - v_2}{35}, \quad i_2 = \frac{1}{10} \left(76 - \frac{3v_2}{7} \right)$$

Nodal analysis (KCL) was more straightforward and efficient.



KCL $\frac{5i_2 - v_2}{45} + 0.02v_1 + \frac{100 - v_2}{30} + i_2 = 0$

$$\Rightarrow 0.2v_3 - 100 = v_1$$

$$v_3 = 50i_2 ; v_1 = 10i_2 - 100 ; v_2 = -40i_2$$

$$\Rightarrow \frac{5i_2 + 40i_2}{45} + \frac{2(10i_2 - 100)}{100} + \frac{100 - (-40i_2)}{30} + i_2 = 0$$

$$\Rightarrow i_2 = \frac{-60}{159} \text{ A}$$

KVL $50i_2 + 30(i_2 - i_3) + 20(i_2 - i_1) = 0$

$$\Rightarrow 2i_1 - 10i_2 + 3i_3 = 0 \quad \text{--- (1)}$$

$$20(i_1 - i_2) + 100 - 0.2v_3 = 0$$

$$\Rightarrow 2i_1 - 3i_2 = -10 \quad \text{--- (2)}$$

$$30(i_3 - i_2) + 45i_4 + 5i_2 - 100 = 0$$

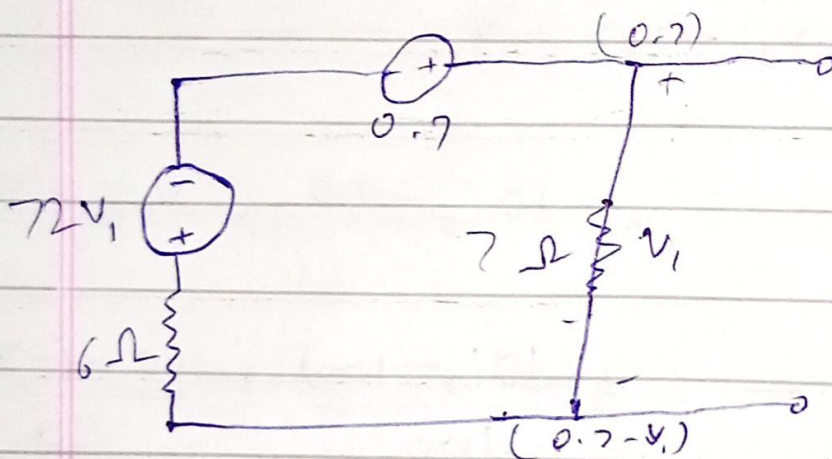
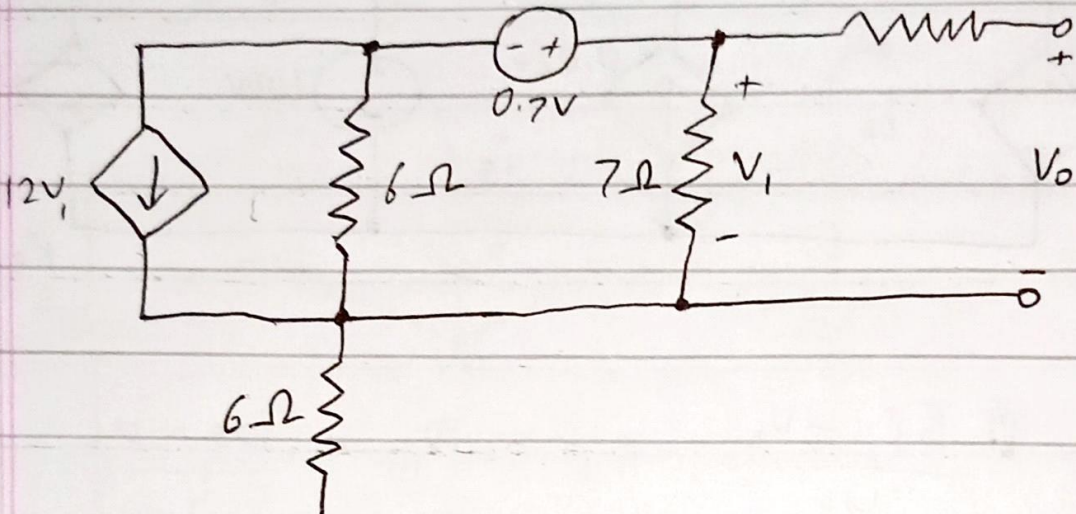
$$\Rightarrow 18i_1 - 43i_2 + 75i_3 = 100 \quad \text{--- (3)}$$

Solving, we get $i_2 = \frac{-60}{159} \text{ A}$

Clearly, nodal analysis (KVL) was simpler.

87. ECA Ch-5.

23)



$$V_0 = V_1$$

$$\frac{V_1}{7} + \frac{0.7 - V_1 - 72V_1}{6} = 0$$

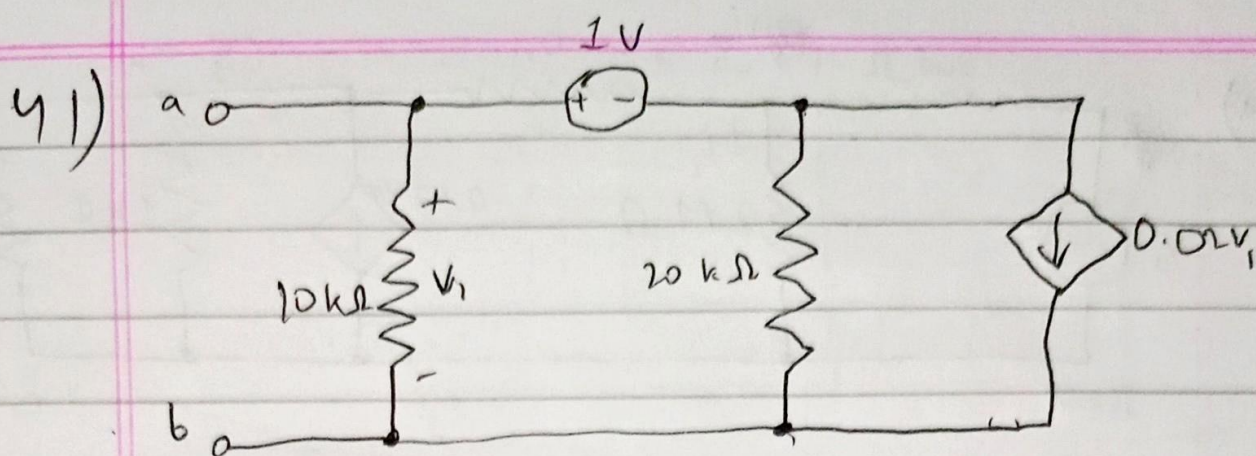
$$\Rightarrow \frac{V_1}{7} + \frac{0.7 - 73V_1}{6} = 0$$

$$\Rightarrow 6V_1 + 4.9 - 511V_1 = 0$$

$$\Rightarrow 505V_1 = 4.9$$

$$\Rightarrow V_1 = \frac{4.9}{505} \text{ V}$$

$$\Rightarrow \boxed{V_1 = 9.7 \text{ mV}} = V_0$$



Nodal Analysis: $\frac{V_1}{10 \times 10^3} + \frac{V_2}{20 \times 10^3} = -0.02V_1$

$$V_2 - V_1 = 1$$

$$\Rightarrow V_1 = \underline{\underline{-2.481 \text{ mV} = V_{th}}}$$

Test current: 1 A.

$$\frac{V_{tc}}{10 \times 10^3} + \frac{V_{tc}}{20 \times 10^3} = 1 - 0.02V_{tc}$$

$$\Rightarrow V_{tc} = 49.63 \text{ V}$$

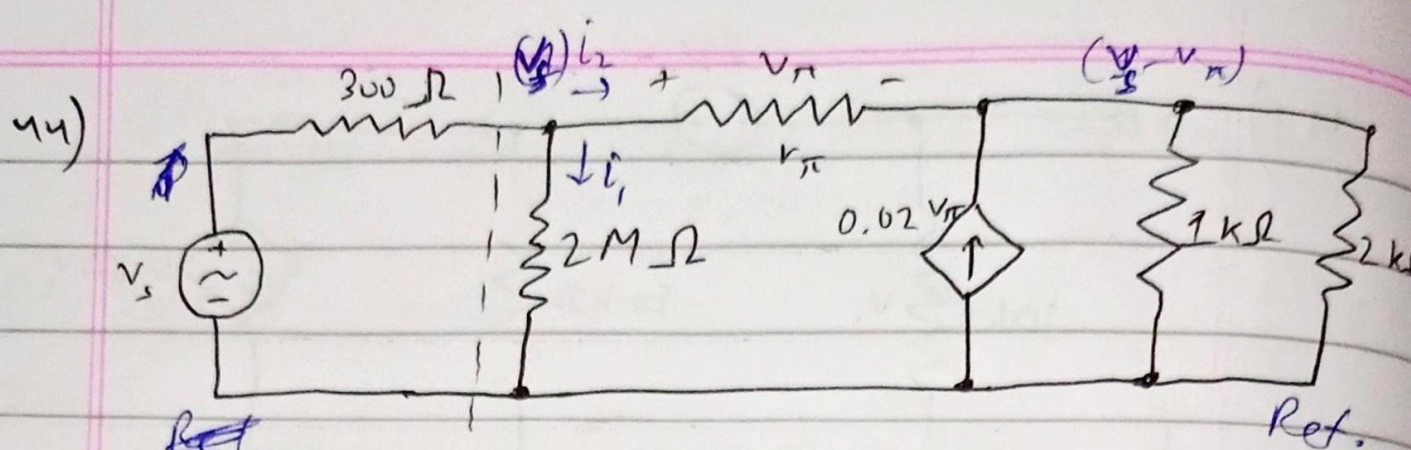
$$\therefore R_{th} = \frac{V_{tc}}{1} = \underline{\underline{49.63 \Omega}}$$

$$\text{Power, } P = \left(\frac{-2.481 \times 10^{-3}}{R + 49.63} \right)^2 \cdot R$$

a) $P_{1k\Omega} = \underline{\underline{5.587 \text{ mW}}}$

b) $P_{4.7k\Omega} = \underline{\underline{1.282 \text{ mW}}}$

c) $P_{10.5k\Omega} = \underline{\underline{578.5 \text{ pW}}}$



Test voltage : $v_s = 1 \text{ V (DC)}$

$$i_1 = \frac{1}{2 \times 10^6} \text{ A}$$

Nodal Analysis:

$$\frac{V_{\pi}}{r_{\pi}} + 0.02 V_{\pi} + \frac{3(V_{\pi} - 1)}{2 \times 10^3} = 0$$

$$V_{\pi} = i_2 r_{\pi}$$

$$i_2 + 0.02 i_2 r_{\pi} + \frac{3}{2 \times 10^3} (i_2 r_{\pi} - 1) = 0$$

$$i_2 + 0.02 i_2 r_{\pi} + \frac{3 i_2 r_{\pi}}{2 \times 10^3} = \frac{3}{2 \times 10^3}$$

$$\left(i_2 - \frac{1}{2 \times 10^6} \right) \left(1 + r_{\pi} (0.02) + \frac{3 r_{\pi}}{2 \times 10^3} \right) = \frac{3}{2 \times 10^3}$$

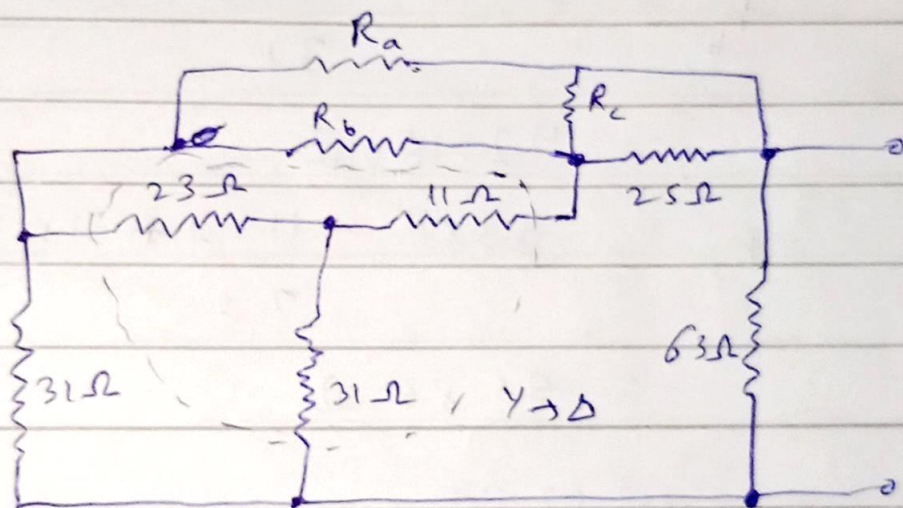
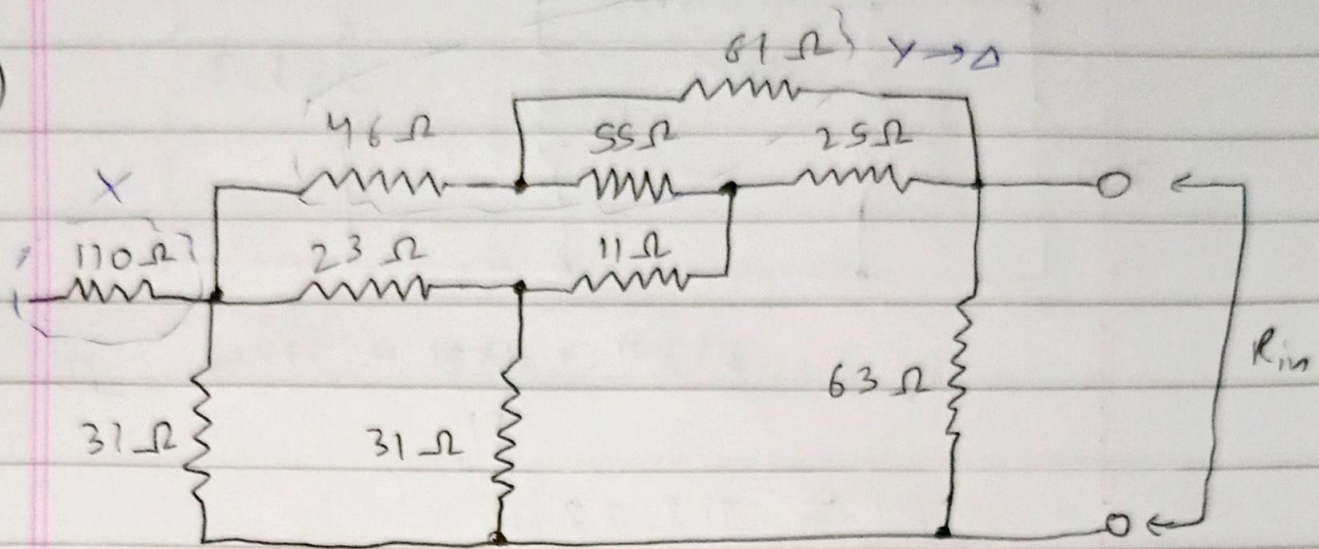
$$\left(i_2 - \frac{1}{2 \times 10^6} \right) (2000 + 43 r_{\pi}) = 3$$

$$i_2 = \frac{3}{2000 + 43 r_{\pi}} + \frac{1}{2 \times 10^6}$$

$$\therefore i_2 = \frac{6 \times 10^6 + 2000 + 43 r_{\pi}}{2 \times 10^6 (2000 + 43 r_{\pi})}$$

$$\Rightarrow R_{TH} = \frac{1}{i} = \frac{2 \times 10^6 (2000 + 43r_{\pi})}{6 \times 10^6 + 2000 + 43r_{\pi}}$$

61)



$$R_a = \frac{46 \times 55 + 55 \times 61 + 61 \times 46}{55} \Omega$$

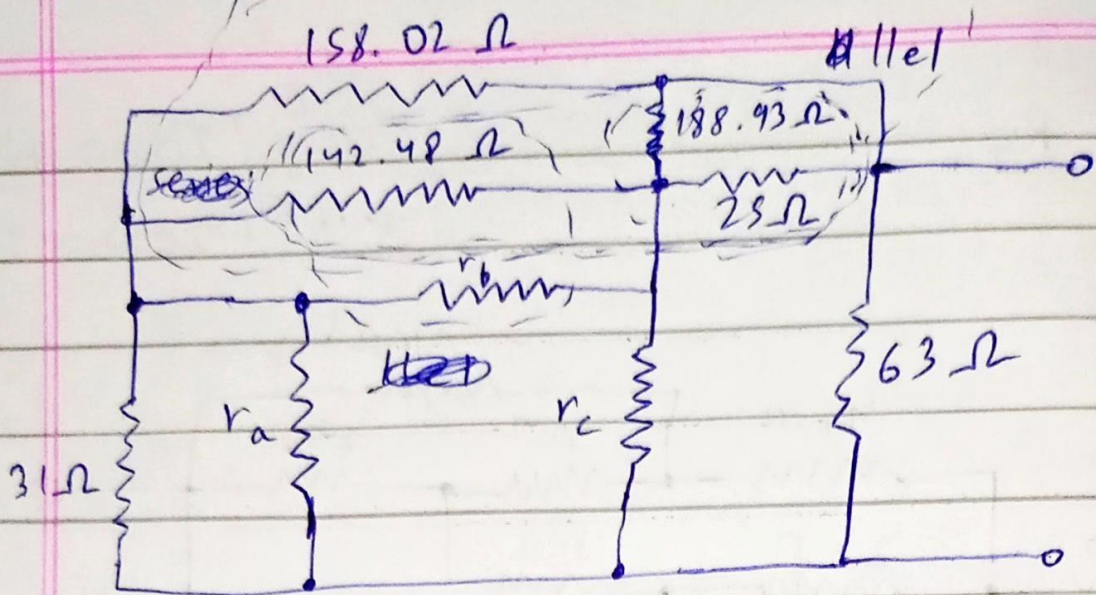
$$= 158.02 \Omega$$

$$R_b = \frac{46 \times 55 + 55 \times 61 + 61 \times 46}{61} \Omega$$

$$= 172.48 \Omega$$

$$R_c = \frac{46 \times 55 + 55 \times 61 + 61 \times 46}{46} \Omega$$

$$= 188.93 \Omega$$



$$r_a = \frac{23 \times 11 + 11 \times 31 + 31 \times 23}{11} \Omega$$

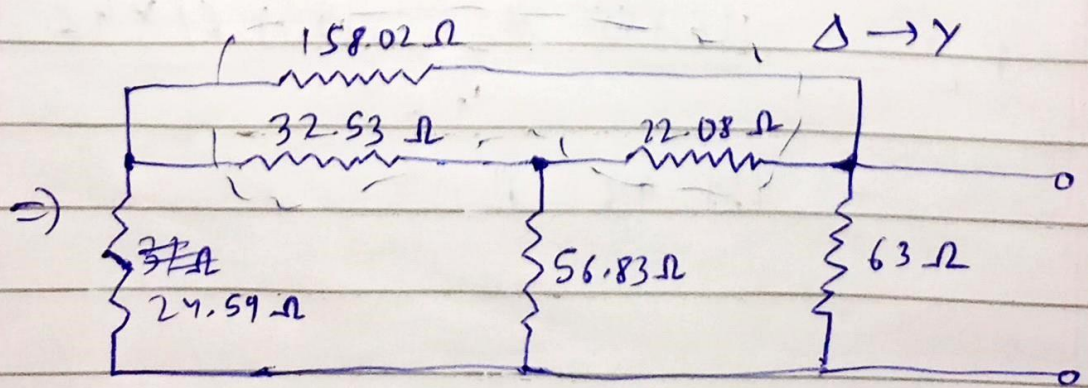
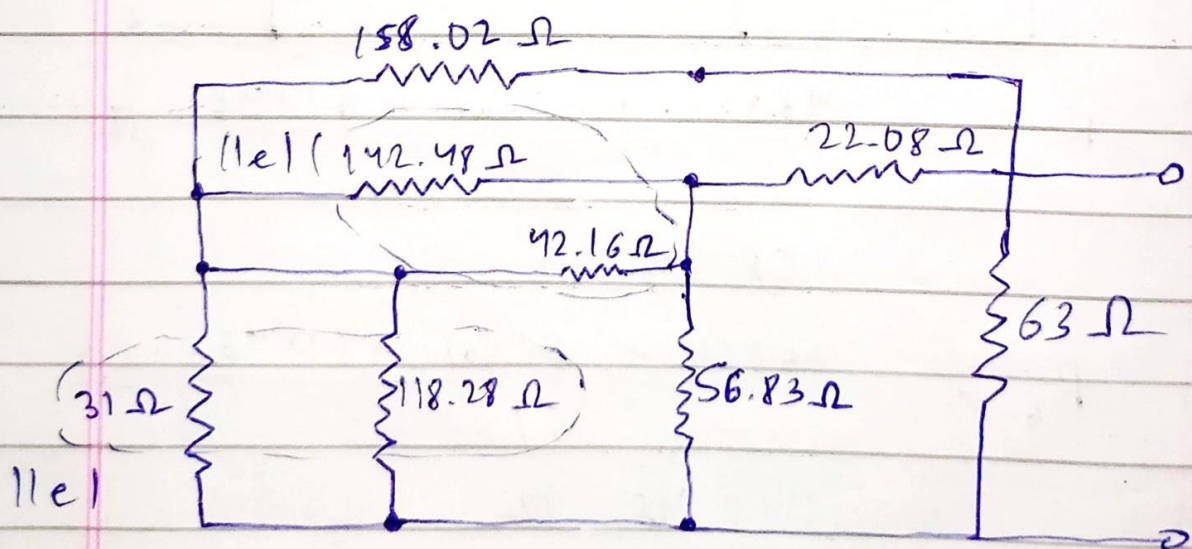
$$= 118.28 \Omega$$

$$r_b = \frac{23 \times 11 + 11 \times 31 + 31 \times 23}{31} \Omega$$

$$= 42.16 \Omega$$

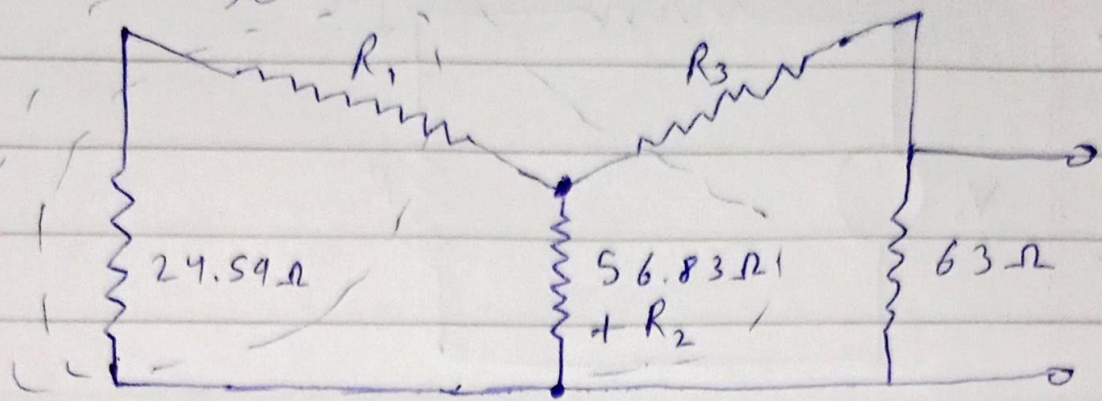
$$r_c = \frac{23 \times 11 + 11 \times 31 + 31 \times 23}{23} \Omega$$

$$= 56.83 \Omega$$



11a)

series



$$R_1 = \frac{158.02 \times 32.53}{158.02 + 32.53 + 22.08} \Omega$$

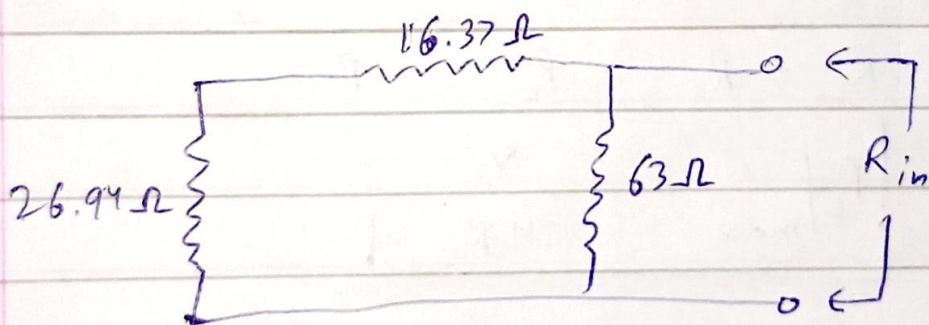
$$= 24.18 \Omega$$

$$R_2 = \frac{32.53 \times 22.08}{158.02 + 32.53 + 22.08} \Omega$$

$$= 3.37 \Omega$$

$$R_3 = \frac{22.08 \times 158.02}{158.02 + 32.53 + 22.08} \Omega$$

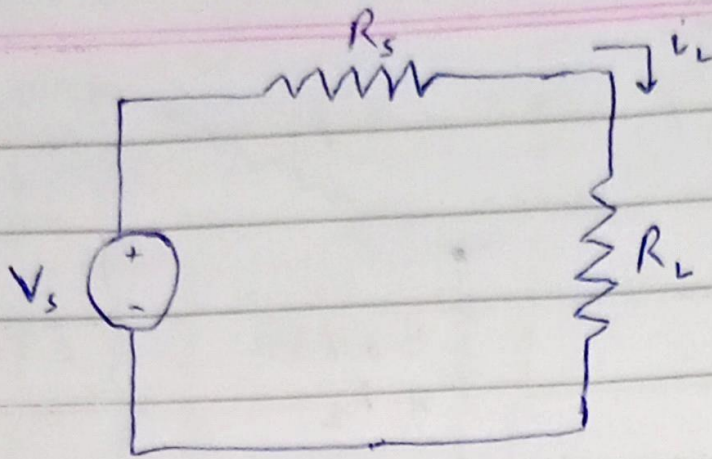
$$= 16.37 \Omega$$



$$\Rightarrow R_{in} = (26.94 + 16.37) \parallel (63)$$

$$\Rightarrow R_{in} = 25.67 \Omega$$

Q8.



Power delivered to load, R_L :

$$P_L = i_L^2 R_L = \frac{V_s^2 R_L}{(R_s + R_L)^2}$$

$$\Rightarrow \frac{dP_L}{dR_L} = \frac{V_s^2 (R_s + R_L)^2 - 2V_s^2 R_L (R_s + R_L)}{(R_s + R_L)^4}$$

For maximum power transfer, $\frac{dP_L}{dR_L} = 0$

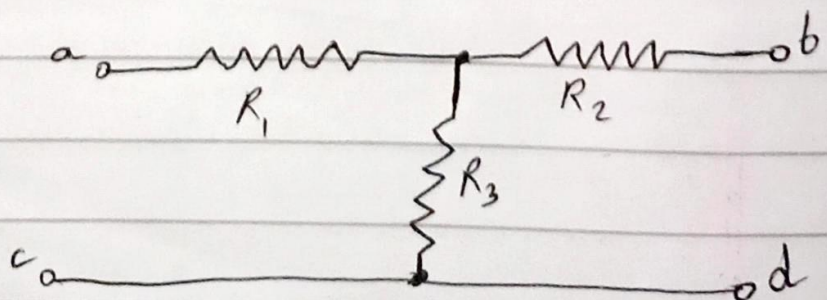
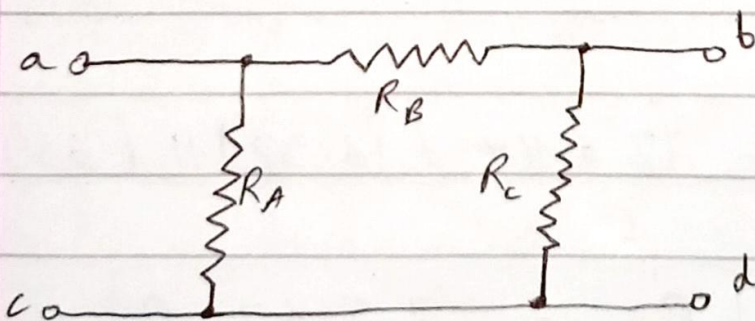
$$\Rightarrow V_s^2 (R_s + R_L) (R_s + R_L - 2R_L) = 0$$

$$\Rightarrow R_s - R_L = 0 \Rightarrow \boxed{R_s = R_L}$$

At $R_L = R_s$, $P_L = P_L|_{\max}$

$$\therefore \boxed{P_L|_{\max} = \frac{V_s^2}{4R_L}}$$

Q9.



$$a) \quad R_{ac} = \frac{(R_B + R_c) \cdot R_A}{R_B + R_c + R_A} = R_1 + R_3$$

$$R_{ab} = \frac{(R_A + R_c) \cdot R_B}{R_A + R_c + R_B} = R_1 + R_2$$

$$R_{bc} = \frac{(R_B + R_A) \cdot R_c}{R_B + R_A + R_c} = R_2 + R_3$$

$$\begin{aligned} \text{Summing up, } R_{ab} + R_{bc} + R_{ca} \\ = \frac{(R_B + R_c) \cdot R_A + (R_A + R_c) \cdot R_B + (R_B + R_A) \cdot R_c}{R_A + R_B + R_c} \end{aligned}$$

$$= R_1 + R_3 + R_1 + R_2 + R_2 + R_3$$

$$\Rightarrow R_1 + R_2 + R_3 = \frac{(R_A + R_B)R_c + (R_B + R_c)R_A + (R_A + R_c)R_B}{2(R_A + R_B + R_c)}$$

$$\text{Now, } R_1 = R_1 + R_2 + R_3 - \underbrace{(R_2 + R_3)}_{R_{bc}}$$

$$\Rightarrow R_1 = \frac{R_A R_B + R_B R_c + R_c R_A}{R_A + R_B + R_c} - \frac{(R_B R_c + R_A R_c)}{R_A + R_B + R_c}$$

$$\Rightarrow R_1 = \frac{R_A R_B}{R_A + R_B + R_c}$$

$$\text{Similarly, } R_2 = \frac{R_B R_c}{R_A + R_B + R_c} \quad \text{and } R_3 = \frac{R_c R_A}{R_A + R_B + R_c}$$

b) Following from (a), consider:

$$R_1 R_2 + R_2 R_3 + R_3 R_1 = \frac{R_A R_B^2 R_C + R_A R_B R_C^2 + R_A^2 R_B R_C}{(R_A + R_B + R_C)^2}$$

$$\Rightarrow \frac{R_A R_B R_C}{R_A + R_B + R_C} = R_1 R_2 + R_2 R_3 + R_3 R_1$$

$$\text{Also, } R_2 = \frac{R_B R_C}{R_A + R_B + R_C}$$

∴ Dividing by R_2 :-

$$\left. \begin{aligned} R_A &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \\ \text{Similarly, } R_B &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \\ R_C &= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \end{aligned} \right\}$$

Q